FIITJEE

JEE (Advanced)-2018

ALL INDIA TEST SERIES

CONCEPT RECAPITULATION TEST - I

Paper 2

Time Allotted: 3 Hours **Maximum Marks: 231**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part is further divided into three sections: **Section-A, Section-C & Section-D**.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

1. ***Section-A (01 – 03, 24 – 26, 47 – 49)*** *contains 9 multiple choice questions which have* ***only one correct answer****. Each question carries* ***+3 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

   Section-A (04 – 08, 27 – 31, 50 – 54) *contains 15 multiple choice questions which have* ***one or more than one correct*** *answer. Each question carries* ***+4 marks*** *for correct answer and* ***–2 marks*** *for wrong answer.*

   Partial Marks **+1** for each correct option provided no incorrect options is selected.

   Section-A (09 – 10, 32 – 33, 55 – 56) *contains 3 paragraphs. Based upon paragraph, 2 multiple choice questions have to be answered. Each question has* ***only one correct*** *answer and carries* ***+3 marks*** *for correct answer. There is no negative marking.*

2. ***Section-C (11 – 20, 34 – 43, 57 – 66)*** *contains 30 Numerical based questions with answer as numerical value from* ***0 to 9*** *and each question carries* ***+3 marks*** *for correct answer. There is no negative marking.*

3. ***Section-D (21 – 23, 44 – 46, 67 – 69)*** *contains 9 Numerical answer type questions with answer XXXXX.XX and each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number		N_a	$= 6.023 \times 10^{23}$
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

PART – I (Physics), PART – II (Chemistry), PART – III (Mathematics):
(SECTION – D)

For questions **21 to 23, 44 to 46, 67 to 69.**
Numerical answer type questions with answer XXXXX. XX

If answer is 348.4 / 251.37 / 213

Correct Method :

0	0	3	4	8	.	4	0
0	0	2	5	1	.	3	7
0	0	2	1	3	.	0	0

Wrong Method :

	3	4	8		.	4	
3	4	8			.		4
		3	4	8	.		4
	3		4	8	.	4	
	2		5	1	.	3	7
		2	1	3	.		
		2	1	3	.	0	
		2	1	3	.		0
		3	4	8	.	4	0
		2	5	1	.	3	7
		2	1	3	.	0	0

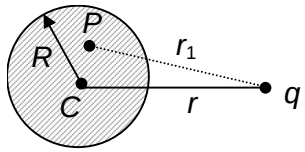
Physics

PART – I

SECTION – A
(One Options Correct Type)

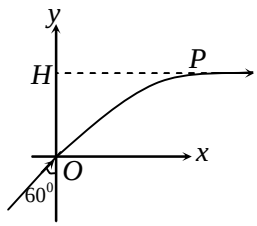
This section contains **3 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

1. A point charge is placed at a distance r from center of a conducting neutral sphere of radius R (r > R). The potential at any point P inside the sphere at a distance r_1 from point charge due to induced charge of the sphere is given by [k = $\frac{1}{4\pi\varepsilon_0}$]



(A) kq/r_1 (B) kq/r
(C) $kq/r - kq/r_1$ (D) $-kq/R$

2. A system of coordinates is drawn in a medium whose refractive index varies as $\mu = \frac{2}{1+y^2}$, where $0 \le y \le 1$. A ray of light is incident at origin at an angle 60^0 with *y* –axis as shown in the figure. At point P ray becomes parallel to x-axis. The value of *H* is



(A) $\left\{\left(\frac{2}{\sqrt{3}}\right)-1\right\}^{1/2}$ (B) $\left\{\frac{2}{\sqrt{3}}\right\}^{1/2}$

(C) $\left\{\left(\sqrt{3}\right)-1\right\}^{1/2}$ (D) $\left\{\left(\sqrt{3}+1\right)\right\}^{1/2}$

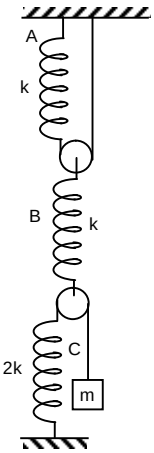
3. An open organ pipe has fundamental frequency 100 Hz. What frequencies will be produced if its one end is closed?
(A) 100,200,300….. (B) 50,150,250,……
(C) 50,100,200,300,…… (D) 50,100,150,200…..

Space for Rough work

(One or More than one correct type)

This section contains **FIVE** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

4. In the given figure, the block is attached with a system of three ideal springs A, B, C. The block is displaced by a small distance x from its equilibrium position vertically downwards and released. T represents the time period of small vertical oscillations of the block (pulleys are ideal). Choose the correct alternative(s).



(A) $T = 2\pi\sqrt{\frac{11m}{2k}}$

(B) the deformation of the spring A is (2/11) times the displacement of the block.

(C) the deformation of the spring C is (1/11) times the displacement of the block.

(D) the deformation of the spring B is (4/11) times the displacement of the block.

5. One mole of monoatomic gas expands with temperature according to the relation $V = KT^{2/3}$ where 'k' is constant. Then choose the correct alternative(s).

(A) Work done to change the temperature by 30°C is 20 R.

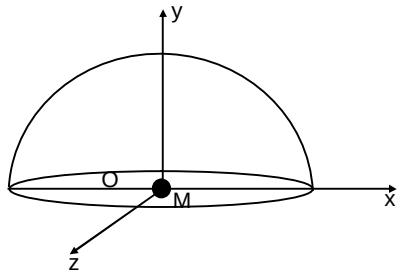
(B) Change in internal energy when temperature is changed by 20°C is 30 R.

(C) If volume of gas is changed by 2% the temperature will change by 3%.

(D) If temperature of gas is changed by 2% the pressure will change by 6%.

Space for Rough work

6. A hemispherical shell of mass 2M and radius 6R and a point mass M are performing circular motion due to their mutual gravitational interaction. Their positions are shown in figure at any moment of time during motion. If r_1 and r_2 are the radii of circular path of hemispherical shell and point mass respectively and ω_1 and ω_2 are the angular speeds of hemi-spherical shell and point mass respectively, then choose the correct statement(s).

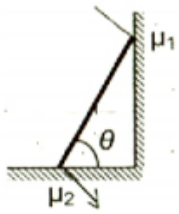


(A) $\frac{r_1}{r_2}=\frac{2}{1}$ (B) $\frac{r_1}{r_2}=\frac{1}{2}$

(C) $\omega_1=\sqrt{\frac{GM}{72R^3}}$ (D) $\omega_2=\sqrt{\frac{GM}{72R^3}}$

7. An ammeter has a resistance of 50 Ω and a full scale deflection current of 50 μA. It can be used as a voltmeter or as a higher range ammeter provided that a resistance is added to it. Choose the correct range and resistance combination from the following
(A) 20 V range with approximately 400 kΩ resistance in series
(B) 100 V range with approximately $2 \times 10^6 \Omega$ resistance in series
(C) 1 mA range with 50 Ω resistance in parallel
(D) 0.1 mA range with 50 Ω resistance in parallel

8. In figure, a ladder of mass m is shown leaning against a wall. It is in static equilibrium making an angle θ with the horizontal floor. The coefficient of friction between the wall and the ladder is μ_1 and that between the floor and the ladder is μ_2. The normal reaction of the wall on the ladder is N_1 and that of the floor is N_2. If the ladder is about to slip, then



(A) $\mu_1=0\ \ \mu_2\neq 0$ and $N_2\tan\theta=\frac{mg}{2}$ (B) $\mu_1\neq 0\ \ \mu_2=0$ and $N_1\tan\theta=\frac{mg}{2}$

(C) $\mu_1\neq 0\ \ \mu_2\neq 0$ and $N_2=\frac{mg}{1+\mu_1\mu_2}$ (D) $\mu_1=0\ \ \mu_2\neq 0$ and $N_1\tan\theta=\frac{mg}{2}$

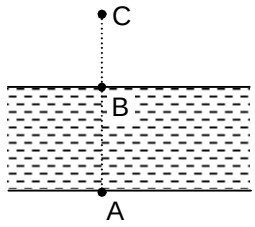
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(Paragraph Type)

This section contains **ONE** paragraph. Based on the paragraph, there are **TWO** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.

Paragraph for Question Nos. 9 and 10

A river of width *d* is flowing with uniform velocity *u*. A boat starts moving from point *A* (one bank of river) with speed *u* relative to the river. The direction of resultant velocity is always perpendicular to line joining boat and fixed point *C* (see figure). Point *B* is on the opposite side of the river and *A*, *B*, *C* are in straight line. If *AB* = *BC* = *d*



9. The path of boat is
   (A) straight line
   (B) parabolic
   (C) elliptical
   (D) circular

10. The time taken by the boat to cross the river is
   (A) $\frac{d}{u}$
   (B) $\frac{d}{u}\ln\left[2+\sqrt{3}\right]$
   (C) $\frac{d}{u}\ln\left[2-\sqrt{3}\right]$
   (D) $\frac{2d}{u}$

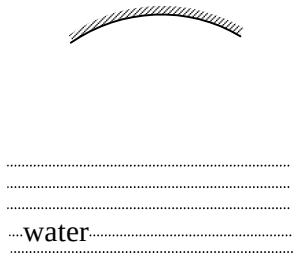
Space for Rough work

SECTION – C
(Single digit integer type)

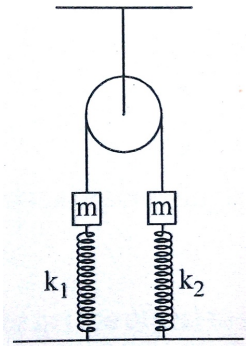
This section contains **TEN** questions. The answer to each question is a single Digit integer ranging from 0 to 9, both inclusive.

11. In YDSE experiment if the screen is shifted by a distance of 0.5 m away from the slit, 3rd maxima is shifted by 3×10^{-4} m. The separation between slit is 2×10^{-3} m. If the wavelength (in 100 nm) used in the experiment is 'k', find the value of 'k'.

12. A concave mirror of focal length 15 cm is placed in air at a height of 45 cm above the water surface as shown. The axis of the mirror is vertical and the reflecting surface faces the water surface. A point object is placed on the axis of concave mirror at a distance 35 cm above the water surface. Find the position of image formed after two successive processes- first reflection from the concave mirror and then first refraction from the water (reflective Index of water 4/3). The distance of image from water surface (in m) is

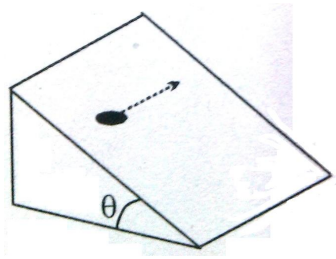


13. On the diagram, two blocks of equal mass are connected by an ideal string. The values of m = 1 kg and k_2 = 100 N/m are given. Initially, both springs are relaxed. Then the left block is slowly pulled down a distance 0.1 m and released. The maximum possible value of k_1 (in N/m) for which both blocks will have same magnitude of acceleration just after releasing, is k. Both spring have natural length of 2m. Then find the value of $\frac{k}{100}$.

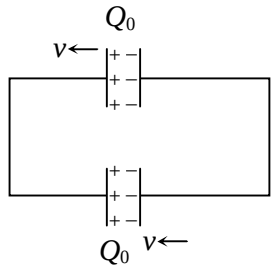


Space for Rough work

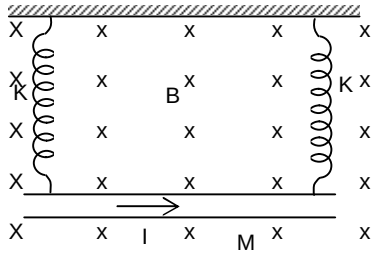
14. An object of mass $m = \sqrt{8}\,kg$ rests on an inclined plane that makes angle $\theta = 45°$ with the horizontal floor. The minimum horizontal force (in N), parallel to the base of the incline that must be applied to the object in order to begin its movement is F. The coefficient of static friction between the object and the plane is $\mu_s = 1.25$. Then the value of $\frac{F}{3}$ is



15. Two identical capacitor connected as shown and having initial charge Q_0. Separation between plates of capacitor is d_0. Suddenly the left plate of upper capacitor and right plate of lower capacitor start moving with speed v towards left while other plate of capacitor remains fixed. (given $\frac{Q_0 v}{2d} = 1$ amp). Find the value of current (in amp) in the circuit.



16. A metal rod of mass 10gm and length 25 cm is suspended on two springs as shown in figure. The springs are extended by 4 cm. When a 20 ampere current passes through the rod it rises by 1 cm. The magnetic field is $x \times 10^{-2}$ T ($g = 10\ m/s^2$). Find the value of 2x.



17. A uniform disc of radius R having charge Q distributed uniformly all over its surface is placed on a smooth horizontal surface. A magnetic field, $B = kxt^2$, where k is a constant, x is the distance (in metre) from the centre of the disc and t is the time (in second), is switched on perpendicular to the plane of the disc. Find the torque (in N-m) acting on the disc after 15 sec. (Take $4kQ = 1$ S.I. unit and $R = 1$ m)

18. Suppose potential energy between electron and proton at separation r is given by $U = k \ln r$, where k is constant. For such hypothetical hydrogen atom, find the ratio of energy difference between energy levels (n = 1 and n = 2) and (n = 2 and n = 4) is

Space for Rough work

19. K_α wavelength emitted by an atom of atomic number Z = 11 is λ. Find the atomic number for an atom that emits K_α radiation with wavelength 4λ.

20. In a certain hypothetical radioactive decay process, species A decays into species B and species B decays into species C according to the reactions

$A \longrightarrow 2B$ + particles + energy

$B \longrightarrow 3C$ + particles + energy

The decay constant for species A is $\lambda_1 = 1 \sec^{-1}$ and that for species B is $\lambda_2 = 100 \sec^{-1}$. Initially 10^4 moles of species of A were present while there was none of B and C. It was found that species B reaches its maximum number at a time t_0 = 2 ln(10)sec. Calculate the value of maximum number of moles of B.

SECTION – D
(Numerical Based XXXXX.XX answer Type)

This section contains **3 questions**. Each question, when worked out will result in numerical answer Type with answer xxxxx.xx.

21. The main scale of vernier calipers reads in mm and its vernier is divided into 10 divisions which coincides with 9 divisions of the main scale. It was also observed when two jaws are brought in contact the zero of the vernier scale coincided with the zero of the main scale. The length of a cube is measured with this vernier calipers. The main scale reads 10 mm and the first division of the vernier scale coincides with the main scale. If the mass of the cube is 2.7369 g then, find the density of the cube in (gm/cm^3) upto appropriate significant figures.

22. A point mass of 1 kg collides elastically with a stationary point mass of 5 kg. After their collision, the 1 kg mass reverses its direction and moves with a speed of 2 ms^{-1}. Find the kinetic energy (in Joule) of a particle of mass 6 kg moving with same velocity as that of centre of mass of the above two particles of mass 1 kg and 5 kg.

23. In YDSE distance between the slits plane and screen is 1m and distance between two slits is 5 mm. If slabs of thickness 2 mm and 1.5 mm having refractive index 1.5 and 1.4 are placed in front of two slits, find the shift (in m) of central maxima

Space for Rough work

Chemistry

PART – II

SECTION – A
(One Options Correct Type)

This section contains **3 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

24. Which of the following solutions of complex compounds have maximum vapour pressure?
(A) $[Co(NH_3)_6]Cl_3$ (B) $[Co(NH_3)_5Cl]Cl_2$
(C) $[Co(NH_3)_4Cl_2]Cl$ (D) $[Co(NH_3)_3Cl_3]$

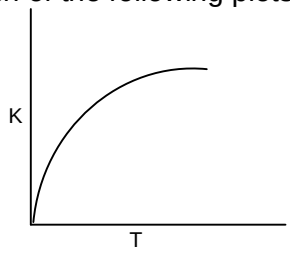
25. Hydrolysis of which of the following compounds is an example of disproportionation reaction.
(A) XeF_6 (B) XeF_4
(C) PCl_5 (D) PCl_3

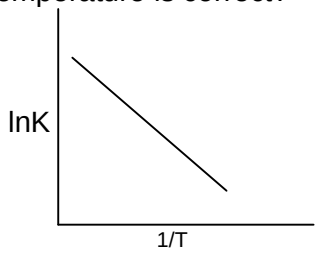
26. Which of the following compounds show π-back bonding?
(A) $AlCl_3$ (B) BF_3
(C) SF_4 (D) BH_3

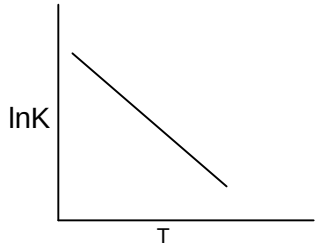
(One or More than one correct type)

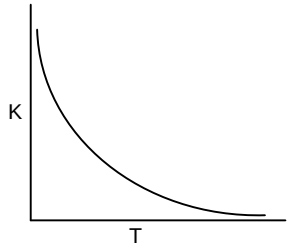
This section contains **FIVE** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

27. Which of the following plots of rate constant vs temperature is correct?

(A) 

(B) 

(C) 

(D) 

Space for Rough work

28. The radial distribution function |P(r)| is used to determine the most probable radius, which is used to find the electron in a given orbital $\frac{dP(r)}{dt}$ for 1s orbital of hydrogen like atom having atomic number Z, is $\frac{dP}{dr}=\frac{4Z^3}{a_0^3}\left(2r-\frac{2Zr^2}{a^0}\right)e^{-2Zr/a_0}$. Then which of the following statements is/are correct?

(A) At the point of maximum value of radial distribution function $\frac{dP(r)}{dr}$ = 0; one antinode is present.

(B) Most probable radius of Li^{2+} is $\frac{a_0}{3}$ pm

(C) Most probable radius of He^{+} is $\frac{a_0}{2}$ pm

(D) Most probable radius of hydrogen atom is a_0 pm

29. The normal boiling point of a liquid X is 400 K. ΔH_{vap} at normal boiling point is 40 kJ/mol. Select the correct statements.

(A) ΔS_{vap} < 100 J/mol K at 400 K and 2 atm

(B) ΔS_{vap} < 10 J/mol K at 400 K and 1 atm

(C) ΔG_{vap} < 0 at 410 K and 1 atm

(D) ΔU = 43.22 kJ/mol K at 400 K and 1 atm

30. Which of the following are concentration cells?

(A) $Pt\,|\,\underset{P_1}{H_2(g)}\,|\,HCl\,|\,\underset{P_2}{H_2(g)}\,|\,Pt$

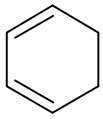
(B) $\underset{a_1}{Cd(Hg)}\,|\,\underset{c}{Cd^{2+}}\,|\,\underset{a_2}{Hg(Cd)}$

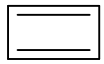
(C) $Zn(s)\,|\,\underset{(C_1)}{Zn^{2+}}\,||\,\underset{(C_2)}{Cu^{2+}}\,|\,Cu$

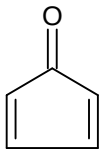
(D) $Ag\,|\,AgCl\,|\,\underset{C_1}{Cl^-(aq)}\,||\,\underset{C_2}{Br^-(aq)}\,|\,AgBr\,|\,Ag$

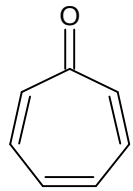
Space for Rough work

31. Which of the following molecule in pure forms is/are unstable at room temperature?

(A) 

(B) 

(C) 

(D) 

(Paragraph Type)

This section contains **ONE** paragraph. Based on the paragraph, there are **TWO** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.

Paragraph for Question Nos. 32 and 33

In complexes of weak field ligands, Δ_0 < P(Pairing energy), the energy difference between t_{2g} and e_g sets is relatively less. Under the influence of strong field ligands Δ_0 > P(Pairing energy), the energy difference between t_{2g} and e_g sets is relatively high.

32. Which of the following is correct statement?
(A) Complex $[Co(H_2O)_6]^{2+}$ is more stable than $[Co(H_2O)_6]^{3+}$
(B) All complexes of Ni(II) are bound to be outer d-orbital complexes
(C) Stability constant for $[Ni(en)_3]^{2+}$ is greater than stability constant for $[Ni(NH_3)_6]^{2+}$
(D) Δ_{oct} for $[Cr(H_2O)_6]^{3+}$ is greater than Δ_{oct} for $[Cr(NH_3)_6]^{3+}$

33. Select the correct increasing order of 10 Dq. value for chromium complexes:
(1) $[Cr(en)_3]^{3+}$ (2) $[Cr(ox)_3]^{3-}$
(3) $[CrF_6]^{3-}$ (4) $[Cr(H_2O)_6]^{3+}$
(A) 4 < 3 < 1 < 2 (B) 3 < 4 < 2 < 1
(C) 4 < 3 < 2 < 1 (D) 3 < 4 < 1 < 2

Space for Rough work

SECTION – C
(Single digit integer type)

This section contains **TEN** questions. The answer to each question is a single Digit integer ranging from 0 to 9, both inclusive.

34. In the following reaction:
$P_4 + xNaOH + yH_2O \longrightarrow PH_3 + 3NaH_2PO_2$
Sum of x + y is

35. Compound x dissociates according to the reaction $2X(g) \rightleftharpoons 2Y(g) + Z(g)$, with degree of dissociation which is small compared to unity, if the expression for α in terms of equilibrium constant K_P and total pressure P is given as $\alpha = \left(\frac{2K_P}{P}\right)^{1/2}$. The value of x is

36. How many of the following polymer are co-polymer:
Nylon-6, Buna-N, Dacron, Buna-S, Teflon, PHBV

37. How many of the following metals can be refined by vapour phase refining?
Zr, Zn, Cd, Hg, Ni, Co, Pt, Fe, Ti

38. If ΔG^o for the half cell MnO_4^- / MnO_2 in an acid solution is -xF, then find the value of x
Given $\left(\text{Given}: E^o_{MnO_4^-/Mn^{2+}} = 1.5\,V; E^o_{MnO_2/Mn^{2+}} = 1.25\,V\right)$

39. A bulb is having an ideal gas at 27°C on heating the bulb to 227°C, 2 litre of gas measured at 227°C is expelled out. The volume of the bulb in litre is

40. Sample of 28 mL of H_2O_2(aq) solution required 10 mL of 0.1 M $KMnO_4$(aq) solution for complete reaction in acidic medium. What is volume strength of H_2O_2?

Space for Rough work

41. The ionic radii of A^+ and B^- are $1.7\,\overset{o}{A}$ and $1.8\,\overset{o}{A}$ respectively. Find the coordination number of A^+.

42. For a reaction $A \rightarrow B$, ΔH = +ve, graph between log K_{eq} and $\frac{1}{T}$ is a straight line of slope $= -\frac{1}{4.606}$. Find ΔH in calories.

43. B_2H_6 has unusual 'x' centre-2 electron bonds. What is x?

SECTION – D
(Numerical Based XXXXX.XX answer Type)

This section contains **3 questions**. Each question, when worked out will result in numerical answer Type with answer xxxxx.xx.

44. Calculate the boiling point of water at 700 mm pressure of Hg. The heat of vapourisation of water in 540 cal g.

45. Calculate the numerical value of resonance energy of N_2O from the following data: (ΔH_f^0 of N_2O = 82 kJ/mol)

$BE_{N \equiv N}$ = 946.2 kJ/mol
$BE_{N = N}$ = 418 kJ/mol
$BE_{O = O}$ = 497 kJ/mol
$BE_{N = O}$ = 605.3 kJ/mol

46. Two buffers(X) and (Y) of pH 4.0 and 6.0 respectively are prepared from acid HA and the salt NaA. Both the buffers are 0.5 M in HA. What would be the pH of the solution obtained by mixing equal volume of two buffers?
$K_{HA} = 1.0 \times 10^{-5}$
log 5.05 = 0.7

Space for Rough work

Mathematics

PART – III

SECTION – A
(One Options Correct Type)

This section contains **3 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

47. If P be a point inside and equilateral triangle ABC such that PA=3, PB=4 and PC=5, then the side length of the equilateral triangle ABC is

(A) $\sqrt{25-12\sqrt{3}}$ (B) 13

(C) $\sqrt{25+12\sqrt{3}}$ (D) 17

48. If f(x) be an invertible function with $f'(x)f''(x) > 0 \forall x \in R$, then $\frac{d^2f^{-1}(x)}{dx^2}$, where $f^{-1}(x)$ is the inverse function of f(x) is

(A) positive $\forall x \in R$

(B) negative $\forall x \in R$

(C) data insufficient

(D) $\frac{d^2f^{-1}(x)}{dx^2}$ will have sign opp. to $\frac{d^2f^{-1}(x)}{dx^2}$

49. For $\alpha \in R$, the range of the function $f(\alpha) = \int_{\tan^{-1}\alpha}^{\cot^{-1}\alpha} \left(\frac{\tan x}{\tan x + \cot x}\right) dx$ is equal to

(A) $\left(\frac{-\pi}{2}, \frac{\pi}{2}\right)$ (B) $(0, \pi)$

(C) $\left(\frac{-\pi}{4}, \frac{3\pi}{4}\right)$ (D) $\left(0, \frac{\pi}{2}\right)$

Space for Rough work

(One or More than one correct type)

This section contains **FIVE** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

50. 3 married couples are to seated around a circular table. Let x denotes the number of ways where atmost two couples sit together and let y denotes the number of ways where only two particular couples sit together, then identify the correct option(s)
(A) $x - 8y = 60$ (B) $x + y = 113$
(C) $x = 13y$ (D) None of these

51. Let a, b, c be unequal real numbers. If a, b, c are in G.P. and $a + b + c = bx,$ then 'x' can not be equal to:
(A) –1 (B) 0
(C) 2 (D) 3

52. Let $f : \left[0, \frac{5}{2}\right) \to \left(\frac{1}{2}, 3\right]$ such that $f(x) = \frac{[x]+1}{\{x\}+1}$ where $[.]$ represents the greatest integer function and $\{.\}$ represents the fractional part of x, then which of the following is / are true.
(A) f(x) is injective discontinuous function.
(B) f(x) is surjective non – differentiable function.
(C) $\min\left(\lim_{x\to 1^-} f(x), \lim_{x\to 1^+} f(x)\right) = f(1)$
(D) $\max\left(\lim_{x\to 2^-} f(x), \lim_{x\to 2^+} f(x)\right) = f(2)$

53. If x, y and z lie on domain of the equation such that ln(x + z) + ln(x – 2y + z) = 2 ln (x – z), then which of the following is / are true
(A) $y = \frac{2xz}{x+z}$ (B) $y^2 = xz$
(C) 2y = x + z (D) $\frac{x}{z} = \frac{x-y}{y-z}$

Space for Rough work

54. The content of 3 bags w.r.t. green and red marbles is as given in the table shown.

Bag	Green	Red
A	3	1
B	2	2
C	1	3

A child randomly selects one of the bags, and draws a marble from it and retains it. If the marble is green, the child draws the second marbles randomly from one of the two remaining bags. If the first marble drawn is red the child draws one more marble from the same bag. The probability that the second drawn marble is green is expressed as $\frac{m}{n}$ (where m and n are coprime). Then

(A) $n - m = 71$ (B) $n - 2m + 2 = 0$

(C) $n - m = 73$ (D) none of the above

(Paragraph Type)

This section contains **ONE** paragraph. Based on the paragraph, there are **TWO** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.

Paragraph for Question Nos. 55 and 56

Let S=0 be the equation of the locus of point P in the plane of a square ABCD such that $\max(PA,PC)=\frac{1}{\sqrt{2}}(PB+PD)$.

55. The ratio of radii of circumcircle of square ABCD and director circle of S = 0 is

(A) $\frac{\sqrt{3}}{2}$ (B) $\frac{1}{4}$

(C) $\frac{1}{2}$ (D) none of these

56. The ratio of perimeter of S=0 and the given square is

(A) $\frac{1}{2\sqrt{2}}$ (B) $\frac{1}{\sqrt{3}}$

(C) $\frac{1}{7}$ (D) $\frac{\pi}{2\sqrt{2}}$

Space for Rough work

SECTION – C
(Single digit integer type)

This section contains **TEN** questions. The answer to each question is a single Digit integer ranging from 0 to 9, both inclusive.

57. If $f:R\to R$ is a function such that $f(x)=x^3+x^2f'(1)+xf''(2)+f'''(3)$ for all $x\in R$, then $f(1)-f(2)$

58. $\left(\sum_{r=1}^{30}\frac{r^2}{\left(r^2-30r+450\right)}\right)-24$ is____

59. Let $f(x)=ax^4+bx^2+3x+7$ and $f(-4)=2286$ and $f(4)=N$. If K be the number of ways in which the number N can be resolved as a product of two divisors which are relatively prime then $K-7$

60. Let two non – collinear vectors $\vec{a}$ and $\vec{b}$ inclined at an angle $\frac{2\pi}{3}$ be such that $|\vec{a}|=3$ and $|\vec{b}|=4$. A point P moves so that any time t the position vector $\overrightarrow{OP}$ (where O is the origin) is given as $\overrightarrow{OP}=\left(e^t+t^{-t}\right)\vec{a}+\left(e^t-e^{-t}\right)\vec{b}$. If the least distance of P from origin is $\sqrt{2}\sqrt{\sqrt{a}-b}$ where $a,b\in N$ then find the value of $\frac{(a+b)}{8}$.

61. Let $A=\left[a_{ij}\right]_{3\times3}$ be a matrix such that $AA^T=4.I$ and $a_{ij}+2c_{ij}=0$ where c_{ij} is the cofactor of $a_{ij}\forall i$ and j and I is the unit matrix of order 3.

If $\begin{vmatrix}a_{11}+4 & a_{12} & a_{13}\\ a_{21} & a_{22}+4 & a_{23}\\ a_{31} & a_{32} & a_{33}+4\end{vmatrix}+5\lambda\begin{vmatrix}a_{11}+1 & a_{12} & a_{13}\\ a_{21} & a_{22}+1 & a_{23}\\ a_{31} & a_{32} & a_{33}+1\end{vmatrix}=0$, then find the value of 5λ is

Space for Rough work

62. Let $f(x)$ be a cubic polynomial which has local maximum at x = –1 and $f'(x)$ has a local minimum at $x=1$. If $f(-1)=10$ and $f(3)=-22$, If the distance between its two horizontal tangents is D then $\frac{D}{8}$ is

63. In ΔABC, circumradius is 3 and inradius is 1.5 units. If the value of $a\cot^2 A + b^2\cot^3 B + c^3\cot^4 C$ is $m\sqrt{n}$ where m and n are prime numbers, then find the value of $|(m-2n)|$.

64. Let a, b, c, d be four distinct real numbers in A.P. Find the smallest positive value of k satisfying $2(a-b)+4k(b-c)^2+(c-a)^3=2(a-d)+(b-d)^2+(c-d)^3$

65. Number of positive integer x such that product of its digits in the decimal representation is $x^2-10x-22$ is

66. The side of a triangle are in the ratio 3:5 and the third side is 16, if the largest possible area of the triangle is k, then $\frac{k}{30}$ is _____

SECTION – D
(Numerical Based XXXXX.XX answer Type)

This section contains **3 questions**. Each question, when worked out will result in numerical answer Type with answer xxxxx.xx.

67. If the dependent variable y is changed to 'z' by the substitution $y=\tan z$ and the differential equation $\frac{d^2y}{dx^2}=1+\frac{2(1+y)}{1+y^2}\left(\frac{dy}{dx}\right)^2$ is changed to $\frac{d^2z}{dx^2}=\cos^2 z+k\left(\frac{dz}{dx}\right)^2$, then the value of $(305.33)\times k$ equals

68. ABC is a right angle triangle (right angled at B) inscribed in the parabola $y^2 = 4x$. The minimum length of the intercept cut off by the tangents at A and C to the parabola from y-axis is λ then 1002.2λ is

69. Rectangle ABCD has area 200. An ellipse with area 200π passes through A and C and has foci at B and D. Let perimeter of the rectangle ABCD is P, then 103.44P is

Space for Rough work